



ALBA Accelerator Physics Course

Lattice Design Workshop

This report summarizes the design and study of FODO and
DBA accelerator lattices using OPA.

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Date: May 2026

Repository: Github-Repo

FODO Lattice

1. FODO Cell Design

A storage ring composed of 36 FODO cells is designed for a 2.5 GeV electron beam. Each FODO cell is built from the following magnetic and drift elements:

- Drift sections (D1) with length $L_{D1} = 1.2$ m.
- Focusing and defocusing quadrupoles (QF and QD) with length $L_{QF,QD} = 0.25$ m and quadrupole strength $k_{QF,QD} = \pm 1.2 \text{ m}^{-2}$.
- One bending magnet (BEND) with length $L_{BEND} = 1.2$ m, bending angle $\theta_{dipole} = 360^\circ/36 = 10^\circ$, and entrance/exit edge angles $T_1 = T_2 = 0.5\theta_{dipole} = 5^\circ$.

The FODO cell is assembled according to the following element sequence shown in Figure 1:

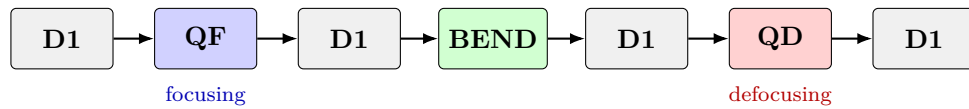


Figure 1: Layout of one FODO cell.

The complete storage-ring lattice is defined by repeating this cell 36 times. Implementing this FODO lattice with the OPA software, a periodic solution is achieved with the linear optics for one cell shown in Figure 2.

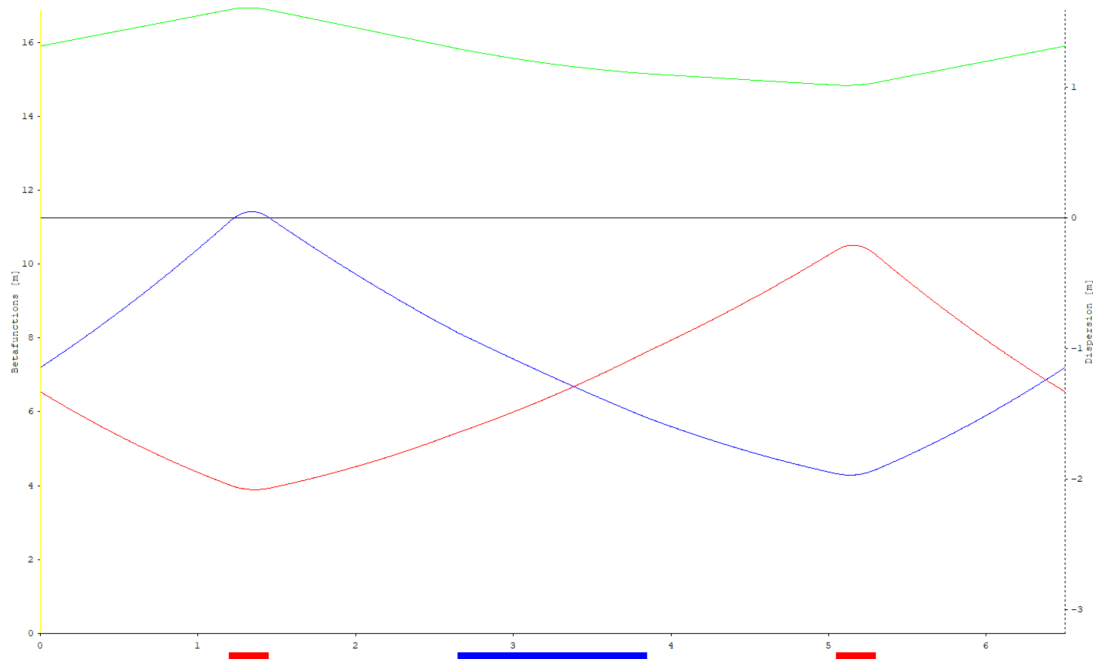


Figure 2: Linear optics of one FODO cell obtained with OPA.

2. Tune dependence and Stability Limits

Quadrupole strength governs how beam particles are focused, impacting the accelerator optics. A focusing quadrupole (QF) focuses horizontally, in the X plane, and defocuses vertically, in the Y plane, by construction, while a defocusing quad (QD) behaves analogously with planes exchanged.

With the FODO cell implemented in OPA, an analysis of how quad strength impacts the tune of the storage-ring is performed. Quad strength and tune dependency is represented in Figure 3. In Fig. 3a an increase of k_{QF} is performed, causing horizontal tune Q_x to increase rapidly and vertical tune Q_y to decrease. In a FODO cell, the tune change is given by

$$\begin{aligned}\Delta Q_x &= \frac{1}{4\pi} \int \beta_x(s) \Delta k(s) ds, \\ \Delta Q_y &= -\frac{1}{4\pi} \int \beta_y(s) \Delta k(s) ds.\end{aligned}\tag{1}$$

and QF is positioned where β_x is large and β_y is small. Therefore ΔQ_x should be larger while ΔQ_y should be smaller for an increase of k_{QF} , which is observed in Fig. 3a. Consequently, the QF is more efficient at changing the tune in the horizontal plane X.

Regarding a change in the defocusing quadrupole strength Δk_{QD} , it is expected to be more efficient at changing tune in the vertical plane Y due to being placed where β_y is large and β_x is small. This corresponds to the results shown in Fig. 3b, where a change in k_{QD} is performed. It is observed that as the strength of the defocusing quad gets larger, in absolute value, the vertical tune (Q_y) increases with similar slope as an increase of k_{QF} relates to horizontal tune (Q_x).

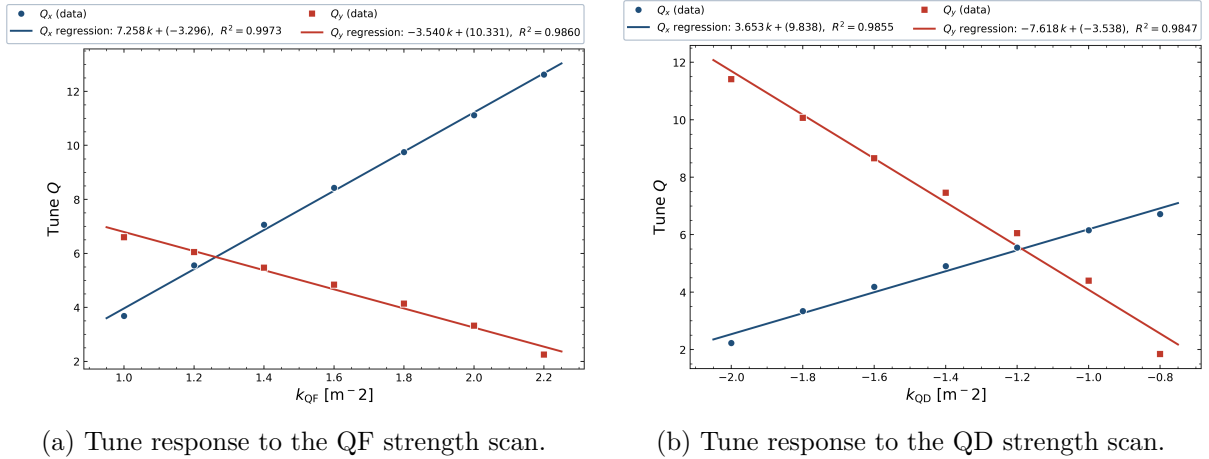


Figure 3: Horizontal and vertical tune dependence on quadrupole strength for the FODO lattice.

3. Independent Horizontal and Vertical Tune Control

An independent change in one transverse tune can be achieved by varying both the focusing and defocusing quadrupole families so that either $\Delta Q_y = 0$ or $\Delta Q_x = 0$ is imposed. To first order,

the required relation between QF and QD is

$$\begin{aligned}\Delta Q_y = 0 &\implies \Delta k_D = -\frac{\beta_y^F}{\beta_y^D} \Delta k_F \\ \Delta Q_x = 0 &\implies \Delta k_F = -\frac{\beta_x^D}{\beta_x^F} \Delta k_D\end{aligned}\tag{2}$$

where β are the beta functions of Twiss in the horizontal and vertical plane at each quadrupole position.

3.1. $\Delta Q_y = 0$ and $\Delta Q_x \neq 0$

Initial tune parameters with the default quadrupole strengths $k_{F,D} = \pm 1.2 \text{ m}^{-2}$ are $Q_x = 5.55289$, $Q_y = 6.05004$ with $\beta_y^F = 3.932$, $\beta_y^D = 10.248$. Using the above relations in Eq. 2, for a change of $\Delta k_F = +0.1 \text{ m}^{-2}$ the expected change in QD for the vertical tune to remain invariant is

$$\Delta k_D^{\text{theory}} = -\frac{3.932}{10.248} \cdot 0.1 = -0.038368 \text{ m}^{-2}$$

Implementing these strength changes in OPA the resulting tunes are $Q_x = 6.22831$, $Q_y = 6.06335$. Hence, the vertical tune does not remain perfectly invariant. The theoretical increase $\Delta k_D^{\text{theory}}$ produces an error of 0.22% in the vertical tune Q_y .

In addition, arbitrary shifts Δk in OPA can be applied to observe its effects in tune. After a change $\Delta k_F = +0.1 \text{ m}^{-2}$ is applied, to keep Q_y invariant, OPA requires a correction in QD of

$$\Delta k_D^{\text{OPA}} = -0.036634 \text{ m}^{-2},$$

which results in the final tunes $Q_x = 6.23295$, $Q_y = 6.05004$. The difference between the theoretical and simulated values of Δk_D is 4.7%.

Table 1: Initial, theoretical, and OPA cases for the condition $\Delta Q_y = 0$. The errors are evaluated with respect to the OPA solution.

Case	$\Delta k_D \text{ (m}^{-2}\text{)}$	Q_x	Q_y	Error in Δk_D	Final tune error
Initial	—	5.55289	6.05004	—	—
Theory	-0.038368	6.22831	6.06335	4.7%	0.22% in Q_y
OPA	-0.036634	6.23295	6.05004	0%	0% in Q_y

Obtained results are presented in a more comprehensive way in Table 1. Given the presented errors, the given formula provides an acceptable approximation for small variations in Δk . The deviation arises because the formula is derived from a first-order perturbative approximation, assuming that the β functions remain constant when k changes. In reality, the β functions also change with the quadrupole strength. Furthermore, the derivation assumes the thin-lens approximation, in which β is considered constant throughout the quadrupole length, whereas OPA does not make these approximations.

3.2. $\Delta Q_y \neq 0$ and $\Delta Q_x = 0$

In an analogous approach as the previous case, for a variation $\Delta k_D = -0.1 \text{ m}^{-2}$ the required change in QF to keep the horizontal tune invariant is

$$\Delta k_F = -\frac{\beta_x^D}{\beta_x^F} \Delta k_D = -\frac{4.435}{11.257} \cdot (-0.1) = 0.039398 \text{ m}^{-2}.$$

Testing this value in OPA gives the results presented in Table 2. Thus, Q_x does not remain completely invariant, introducing an error in horizontal tune of 0.34%. This can be considered an acceptable approximation up to the second decimal of the tune. According to the OPA simulation, the actual variation required in the focusing quadrupole for Q_x to remain invariant is $\Delta k_F^{\text{OPA}} = 0.037164 \text{ m}^{-2}$. Therefore, the theoretical approximation has an error of 6% in the needed quadrupole strength change.

Table 2: Initial, theoretical, and OPA cases for the condition $\Delta Q_x = 0$. The errors are evaluated with respect to the OPA solution.

Case	$\Delta k_F \text{ (m}^{-2}\text{)}$	Q_x	Q_y	Error in Δk_F	Final tune error
Initial	—	5.55289	6.05004	—	—
Theory	0.039398	5.57150	6.67621	6%	0.34% in Q_x
OPA	0.037164	5.55289	6.68173	0%	0% in Q_x

4. Minimum Horizontal Emittance

The equilibrium emittance is determined by the balance between quantum excitation, caused by the discrete emission of synchrotron radiation photons, and radiation damping. The resulting equilibrium emittance can be written as

$$\varepsilon_x = C_q \gamma^2 \frac{I_5}{J_x I_2}$$

which can be approximated to

$$\varepsilon_0 \approx C_q \gamma^2 \theta^3 F \quad (3)$$

where $C_q = 3.84 \times 10^{-13}$, $\gamma = E/m_e c^2 = 4893$ is the Lorentz factor of the beam, $\theta = 10^\circ$ is the bending angle of each dipole magnet, and F is a factor that measures the dependence of the emittance on the lattice style. For a FODO lattice, F depends on its horizontal phase advance. Specifically, for a symmetric FODO cell ($|k_{QF}| = |k_{QD}|$), the minimum value is $F = 1.2$ for a 137° phase advance. This corresponds to a predicted minimum emittance of $\varepsilon_0^{\text{sym}} \approx 58.7 \text{ nm rad}$. Implementing this symmetric case in OPA, a phase advance of $\approx 137^\circ$ is achieved with strength $k_{QF} = 2.4$, $k_{QD} = -2.4$, which yields a horizontal emittance of $\varepsilon = 25.563 \text{ nm rad}$, roughly half the expected value. This discrepancy may be explained by the several approximations that Eq. 3 uses, mainly the thin-lens, small dipole angle and smooth beta function approximations, but also the assumption that there is no gradient dipole component and $j_x \approx 1$. Because of this, Eq. 3 can be interpreted as a scaling law that approximates the order of magnitude of the emittance

and is useful as a reference of how emittance scales when changing the ring geometry. As an example, from Eq. 3 it is trivial to predict that for a FODO lattice the emittance can be lowered increasing the number of cells in the lattice, therefore reducing the bending angle of each dipole. Another approach is manually optimizing the quadrupole strength in OPA to achieve the minimum emittance in the more general case where $|k_{QF}| \neq |k_{QD}|$. With this approach a horizontal emittance of $\varepsilon = 19.905 \text{ nm rad}$ is obtained when $k_{QF} = 2.41$, $k_{QD} = -1.295$, subject to the constraint that the maximum beta functions in both planes do not exceed 25 m. For this optimized lattice, Eq. 3 would have an effective factor $F \approx 0.4$. Additionally, in this configuration the tunes are $Q_x = 14.55501$ and $Q_y = 2.93728$. The FODO optics are presented in Figure 4.

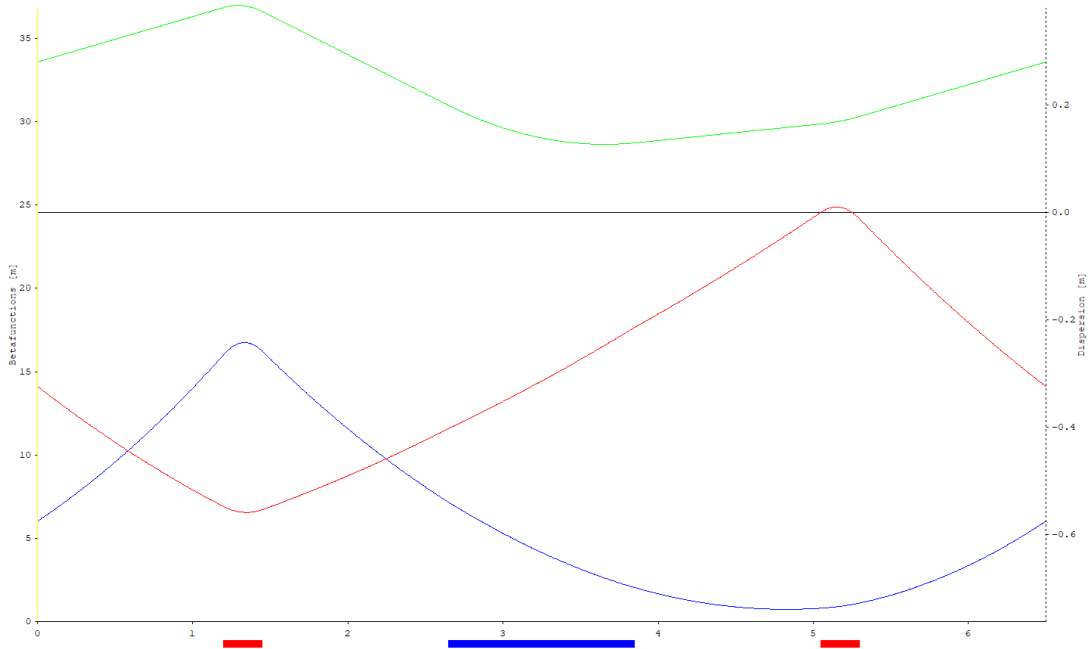


Figure 4: OPA optics for the optimized FODO lattice with minimum horizontal emittance and $\beta_{x,y} < 25 \text{ m}$ constraint.

Given this optimized lattice, the transverse beam envelopes are computed assuming a 2% vertical-horizontal emittance coupling, $\varepsilon_y = 0.02 \varepsilon_x$. Figure 5 shows the resulting FODO envelopes along one FODO cell. The horizontal, in blue, and the vertical, in red, envelopes are given by

$$\sigma_{x,y}(s) = \sqrt{\beta_{x,y}(s) \varepsilon_{x,y}}. \quad (4)$$

The introduced coupling leads to a small vertical beam size with a maximum around $\sigma_y \approx 0.09 \text{ mm}$. The horizontal beam size presents a maximum value of $\sigma_x \approx 0.65 \text{ mm}$, which is well within the beam pipe aperture, guaranteeing an adequate safety margin, i.e. no significant particle losses due to beam interception with the vacuum chamber.

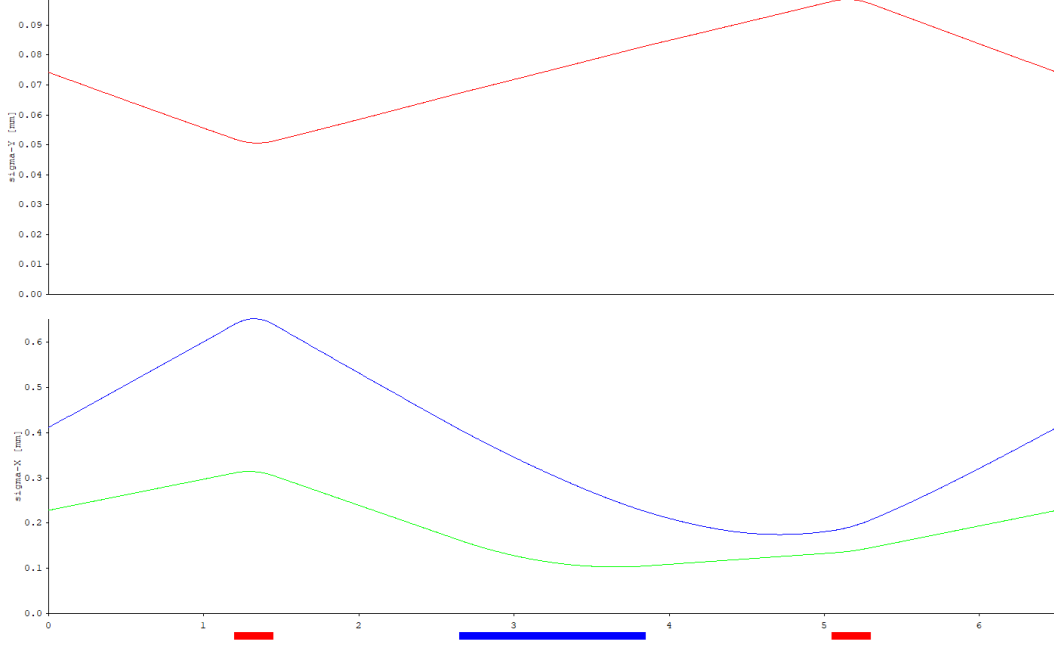


Figure 5: Horizontal and vertical beam envelopes for the optimized FODO lattice, assuming a 2% vertical-horizontal emittance coupling.

5. Impact of Ring Periodicity on Beam Optics

Fixing the optimal quadrupole strengths $k_{QF} = 2.41$ and $k_{QD} = -1.295$, different number of cells for the storage ring are tested. Results presented in Table 3 show the found parameters using OPA with three different number of cells.

The maximum values for β_x remain almost constant, slightly increasing with the number of cells. This follows the horizontal transfer matrix of rectangular dipoles

$$M_{H,\text{rect}} = \begin{pmatrix} 1 & \rho\theta \\ 0 & 1 \end{pmatrix}$$

$$M_{\text{drift}} = \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix}$$

In the horizontal plane, the dipole behaves like a drift, not having any effect in the focusing. Additionally, the slight increase in β_x is explained by the geometric curve particles follow when passing through the dipole. The M_{12} shown in the transfer matrix is an approximation, when considering the full expansion $\rho \sin(\theta) = \rho\theta - \frac{\rho\theta^3}{6} + \dots$ the effective length of the horizontal drift increases slightly with a smaller bending angle θ . Therefore, with more cells a slightly longer drift is expected and therefore a slightly larger β_x .

The vertical beta function β_y increases significantly with the number of cells. Rectangular dipoles have a focusing term in their transfer matrix

$$M_{V,\text{rect}} = \begin{pmatrix} 1 & \rho\theta \\ -\frac{\theta}{\rho} & 1 \end{pmatrix}$$

where the element $M_{21} = -\theta/\rho$ corresponds to a converging thin lens with effective focal length

$$\frac{1}{f_{b,V}} = \frac{\theta}{\rho} = \frac{\theta^2}{L_b}$$

As the number of cells increases with fixed dipole length, the bending angle per dipole decreases as $\theta = 2\pi/N_c$, and therefore $f_{b,V} \propto N_c^2$. Consequently $\beta_{y,\max}$ increases when the number of cells and the focusing distance increase.

Regarding the horizontal dispersion, in Table 3 it is observed $D_{\max}$ presents an approximately inverse linear relation with the cell number, an increase in N_c produces an equivalent decrease in $D_{\max}$. The horizontal dispersion satisfies

$$D''(s) + K(s)D(s) = \frac{1}{\rho(s)}$$

with $1/\rho \propto \theta \propto 1/N_c$. A direct way of explaining this relation is with the 3×3 matrix for a rectangular dipole

$$M_{H,\text{rect}} = \begin{pmatrix} 1 & \rho\theta & \frac{1}{2}\rho\theta^2 \\ 0 & 1 & \theta \\ 0 & 0 & 1 \end{pmatrix}$$

where the component $M_{23} = \theta \propto 1/N_c$ determines the dispersion and therefore it is inversely proportional to the number of cells.

The OPA simulation shows that emittance decreases with the number of cells, specifically with an approximate dependence of $\varepsilon_x \propto 1/N_c^3$. This follows the equilibrium emittance described by Eq. 3 considering $\theta = 360^\circ/N_c$.

Table 3: Effect of storage-ring periodicity on optical parameters.

N_c cells	Bend angle	$\beta_{x,\max}$ (m)	$\beta_{y,\max}$ (m)	$D_{\max}$ (m)	ε_x (nm rad)
18	20°	15.871	13.852	0.7682	160.282
36	10°	16.210	24.575	0.3809	18.905
72	5°	16.298	36.135	0.1900	2.484

DBA Lattice

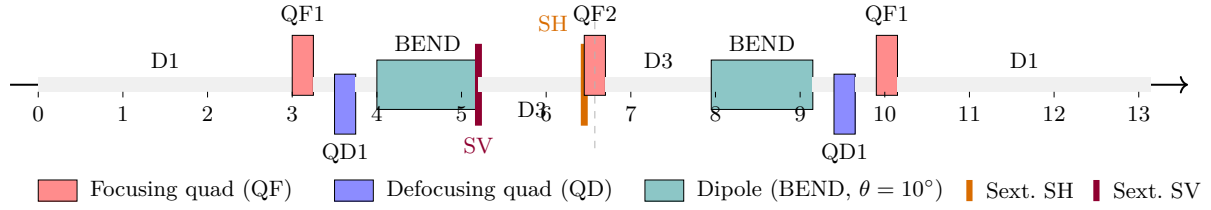


Figure 6: Layout of one DBA cell. Red: focusing quadrupoles (QF1, QF2); blue: defocusing quadrupole (QD1); teal: sector dipoles (BEND, $L = 1.2\text{ m}$, $\theta = 10^\circ$); orange and purple: thin chromaticity-correction sextupoles SH and SV. The ring is composed of 18 such cells.

6. DBA Cell: Linear Optics

6.1. Dispersion Suppression: QF2 Quadrupole Tuning

The achromatic condition for a DBA means a particle that enters the cell with no dispersion ($D_x, D'_x = 0$) must exit without. Using the transfer matrix for a quadrupole, drifts and rectangular dipoles in the thin lens approximation it is found the QF2 focal length needs to satisfy the condition $f = (L_{bend}/2 + L_{drift} + L_{quad}/2)/2$. Then, for this this configuration the theoretical QF2 strength value is

$$k_{QF2}^{theory} = \frac{1}{fL_{quad}} = 4.05\text{m}^{-2}$$

Testing this result in OPA it is found that it does not perfectly suppress dispersion at the ends of the DBA cell, going to $D_x = 0.3\text{m}$.

Adjusting by hand this result, the dispersion at the ends of the cell is exactly zero for when

$$k_{QF2}^{OPA} = 4.23547\text{m}^{-2}$$

Therefore, the theoretical prediction has an error 4%, which is reasonable due to the approximations assumed like thin lens or small angles. The linear optics for this DBA cell are represented in Figure 7, where the correct dispersion suppression is displayed.

6.2. Emittance Optimization

Using the two QF1 and QD1 quadrupoles the horizontal emittance can be reduced to $\varepsilon_x = 5.511\text{ nm rad}$ when quad strengths are $k_{QF1} = 4.305\text{m}^{-2}$ and $k_{QD1} = -4.225\text{m}^{-2}$. In Figure 7, the linear optics for the DBA cell with optimized betas and dispersion is represented. It can also be observed how the maximum beta in both planes is lower than 25m . All the data for this DBA cell configuration is presented in Table 4.

Comparing this results with the FODO ring, for the same energy and number of dipoles a much smaller emittance is found with the DBA ring, roughly a fourth of the previous minimum emittance found with the FODO ring. Consider again the approximation for the equilibrium

emittance

$$\varepsilon_0 \approx C_q \gamma^2 \theta^3 F \quad (5)$$

In the FODO case, the theoretical minimum emittance was found for a 137° FODO, with a factor $F = 1.2$. Additionally, the effective factor that was found when manually optimizing the cell in OPA was $F \approx 0.4$. With Eq. 5, for a DBA lattice style the factor becomes $F = 1/(4\sqrt{15}) = 0.0645$, much smaller than in the FODO case. With this factor, Eq. 5 predicts an equilibrium emittance $\varepsilon^{theory} = 3.16 \text{ nm} \cdot \text{rad}$. Therefore, the obtained results confirm the general theoretical prediction of a DBA cell being capable of obtaining smaller a emittance, although the specific values given by the theoretical approximation are not perfectly accurate.

Table 4: Optics parameters for the minimum horizontal emittance configuration in a DBA lattice.

Parameter	Symbol	Value
Minimum horizontal emittance	ε_x	5.511 nm·rad
Maximum horizontal beta	$\hat{\beta}_x$	24.098 m
Maximum vertical beta	$\hat{\beta}_y$	15.838 m
Maximum horizontal dispersion	D_x	0.3232 m
Horizontal tune	Q_x	17.489
Vertical tune	Q_y	7.701
Horizontal chromaticity	ξ_x	-79.588
Vertical chromaticity	ξ_y	-23.068
<i>Quadrupole strengths</i>		
QF1	k_{QF1}	+4.305 m ⁻²
QD1	k_{QD1}	-4.225 m ⁻²
QF2	k_{QF2}	+4.235 m ⁻²

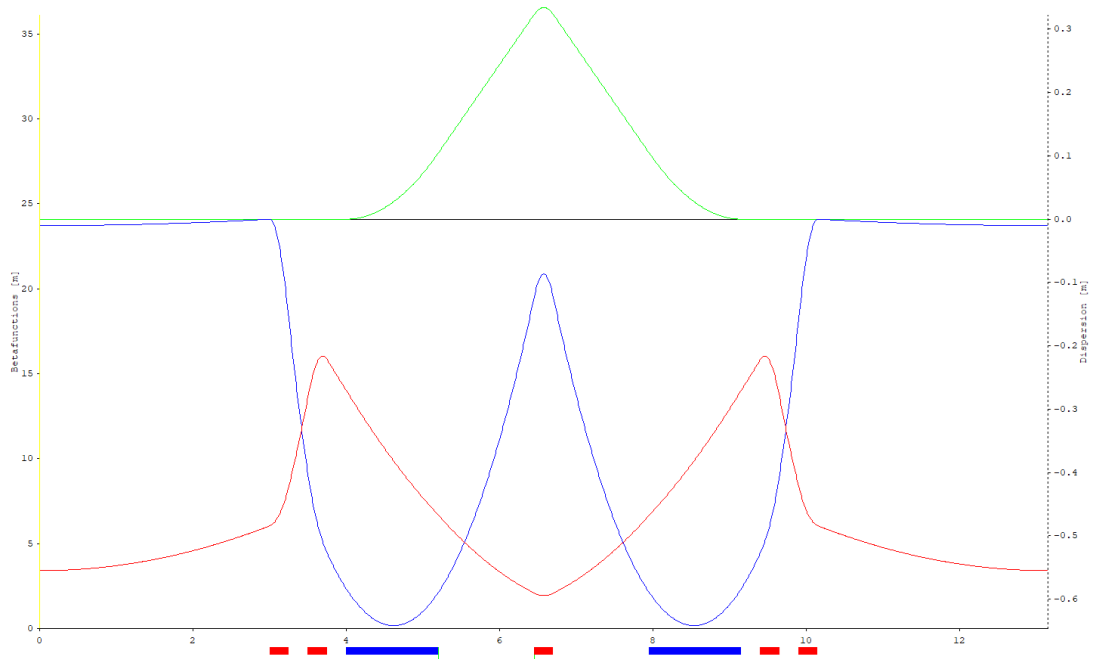


Figure 7: Linear optics of optimized DBA cell obtained with OPA.

6.3. Optimal Sextupole Positions for Chromaticity Correction

Chromaticity can be corrected using sextupoles because they add a focusing component depending on the momentum. The chromaticity correction force of a sextupole is proportional to

$$\Delta\xi_x \propto \beta_x \cdot D_x \cdot mL$$

$$\Delta\xi_y \propto \beta_y \cdot D_x \cdot mL$$

where D_x is the dispersion and m is the sextupole force. Therefore, sextupoles must be placed where the dispersion is not zero $\eta_x \neq 0$ and the beta functions are large. Then, for the focusing sextupole SF, the best position in a DBA cell is near QF2, where β_x is large and the dispersion is the maximum. The defocusing sextupole SD, that corrects ξ_y , needs to be placed where β_y is high and the dispersion is not zero, so the best position would be just after the first dipole or just before the second one. In Figure 6, a layout of the DBA cell is presented where the sextupoles are placed in their optimal positions.

7. DBA Cell: Non-Linear Optics

7.1. Natural Chromaticity

The natural chromaticity of the ring in absence of sextupoles can be calculated for discrete quadrupoles as

$$\xi_{x,y} = \mp \frac{1}{4\pi} \sum_{i \in \text{quads}} \left(k_i \cdot L_i \cdot \beta_{x,y}^{(i)} \right) \quad (6)$$

where k_i is the quad strength, L_i is the quad length and $\beta_{x,y}^{(i)}$ are the beta functions at the (i) -th quad position. For the studied DBA cell, the data needed in the calculation of the natural chromaticity is presented in Table 5. Since the ring is composed of 18 equivalent DBA cells, the total natural chromaticity is obtained by summing the contribution of one cell over the whole ring:

$$\xi_{x,y}^{nat} = \mp \frac{N}{4\pi} \sum_{j=1}^5 k_j L_j \beta_{x,y}^{(j)} = \begin{cases} \xi_x^{nat} = -74.35 \\ \xi_y^{nat} = -19.19 \end{cases}$$

Table 5: Quadrupole parameters measured with OPA for the natural chromaticity calculation.

Index	Element	k [m ⁻²]	L [m]	β_x [m]	β_y [m]
1	QF1	4.30500	0.25	24.098	6.049
2	QD1	-4.22500	0.25	9.084	13.628
3	QF2	4.23547	0.25	19.511	2.082
4	QD1	-4.22500	0.25	4.460	15.838
5	QF1	4.30500	0.25	18.230	8.372

The natural chromaticity computed by OPA for this DBA configuration is

$$\begin{cases} \xi_x^{nat,OPA} = -79.588 \\ \xi_y^{nat,OPA} = -23.068 \end{cases} \quad (7)$$

Then, the values found with the analytical formula Eq. 6 shows an error of 6.5% for the horizontal chromaticity ξ_x^{nat} and a 17% error for the vertical chromaticity ξ_y^{nat} . Thus, the theoretical approach returns a broad approximation at first order, especially in the vertical plane. The differences between the measured natural chromaticity and the theoretical prediction can be attributed to mainly two factors. Firstly, the dipoles also contribute with a quadrupolar component because of their gradient, especially in the vertical plane. Secondly, the thin-lens approximation is not assumed in OPA, the beta functions are numerically integrated along the whole element. Additionally, the larger relative discrepancy in ξ_y arises from a near cancellation between the positive contributions of QF and the negatives from QD. Since the result is the difference of two large numbers of opposite sign, even small relative errors in each term are amplified significantly in the final result.

7.2. Particle Tracking

The tracking of particles is implemented for 1024 turns with initial horizontal amplitudes in the range $x = (0, 1, 2, \dots, 15)$ and $x' = 0$. The result is presented in Figure 8, where the trajectories in the phase space (x, x') form closed concentric ellipses. This shows the linear and stable movement expected by the linear optics of the quadrupoles, with no visible deformations. All particles survive up to initial amplitudes of 15mm, which coincides with the physical aperture of the void camera. Therefore, in absence of sextupoles, the horizontal dynamic aperture equals the physical aperture of the ring.

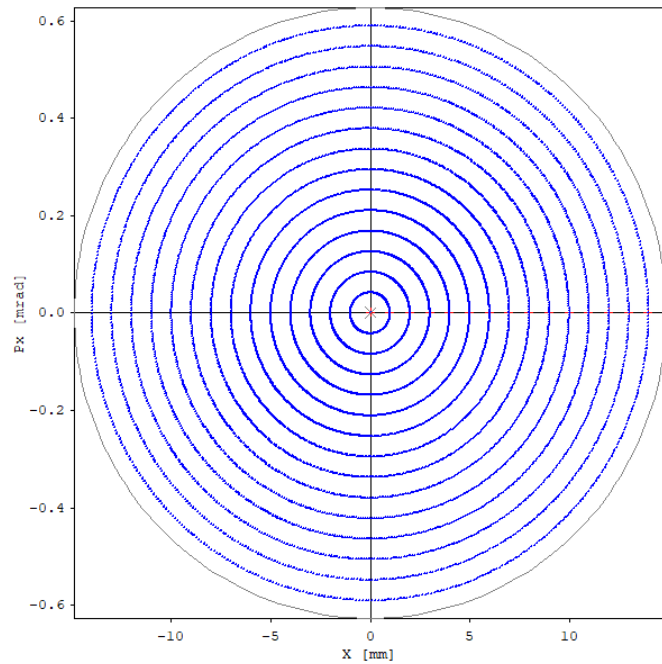


Figure 8: Horizontal phase-space tracking for the DBA lattice.

7.3. Sextupole Strength for Chromaticity Correction

Two sextupoles SH and SV are installed in the DBA cell as is displayed in Figure 6. Previously, it has been observed how quadrupoles generate negative natural chromaticity. Then, for correcting it to zero, the necessary sextupole strengths can be calculated in the thin-lens ($L=0$) approach as the discrete sum

$$\begin{cases} \Delta\xi_x = \frac{1}{4\pi} \sum_i \beta_{x,i} D_i m_i \\ \Delta\xi_y = -\frac{1}{4\pi} \sum_i \beta_{y,i} D_i m_i \end{cases} \quad (8)$$

which is the chromaticity correction given by the sextupoles. Equaling this correction to the natural chromaticity, it can be corrected to zero.

$$\begin{cases} \xi_x^{\text{nat}} + \frac{18}{4\pi} (\beta_{x,SH} D_{SH} m_{SH} + \beta_{x,SV} D_{SV} m_{SV}) = 0 \\ \xi_y^{\text{nat}} - \frac{18}{4\pi} (\beta_{y,SH} D_{SH} m_{SH} + \beta_{y,SV} D_{SV} m_{SV}) = 0 \end{cases} \quad (9)$$

Using the data shown in Table 6 to correct the chromaticity found with OPA in Eq. 7, the necessary strengths are also presented in Table 6.

Table 6: Optical functions at the sextupole positions and theoretical strengths required to correct the natural chromaticity.

Element	β_x [m]	β_y [m]	D_x [m]	m^{theory} [m^{-2}]
SV	2.123	6.665	0.1045	-32.748
SH	19.511	2.082	0.3232	9.963

7.4. Verification of Corrected Chromaticity in OPA

Introducing the theoretical sextupole strengths of m_{SH} and m_{SV} into OPA, taking into account the factor $1/2$ included in the OPA definition of the sextupole strength, the resulting chromaticities are

$$\xi_x \approx -0.02, \quad \xi_y \approx -0.02$$

These residual values are sufficiently close to zero, indicating that the chromaticity correction is effectively achieved. The small remaining deviation can be attributed to numerical rounding effects, discretization of the lattice, and the internal implementation of sextupole kicks in OPA. For comparison, the strengths that provide ideal cancellation of chromaticity in OPA are $m_{SH} = 4.983$ and $m_{SV} = -16.385$. Considering the $1/2$ factor, these results are equivalent to an error of less than 1%, which represents a very good approximation in the thin lens approach. Therefore, the chromaticity correction scheme using two sextupole families located in regions of non-zero dispersion is successfully validated.

7.5. Particle Tracking with Non-Linear Effects

The new ring with sextupoles generates the phase-space trajectories shown in Figure 9, with initial horizontal amplitudes in the range $x = (0, 0.2, 0.4, 0.6, 0.8, 1, 1.1)$ and $x' = 0$. Firstly, it is clearly observed how trajectories stop being ideal ellipses and become distorted by the non-linearity of

the system. This non-linear behaviour leads to the emergence of resonance structures and an amplitude dependence of the betatron tune, meaning that particles with different oscillation amplitudes experience slightly different effective focusing strengths. Therefore, particles with large initial amplitudes exhibit unstable motion and are lost after a finite number of turns, indicating that the dynamic aperture is significantly reduced due to sextupole-induced non-linearities. Consequently, these effects reduce the stability region in phase space, defining a smaller dynamic aperture as is seen in Figure 9 compared to the linear case shown in Figure 8.

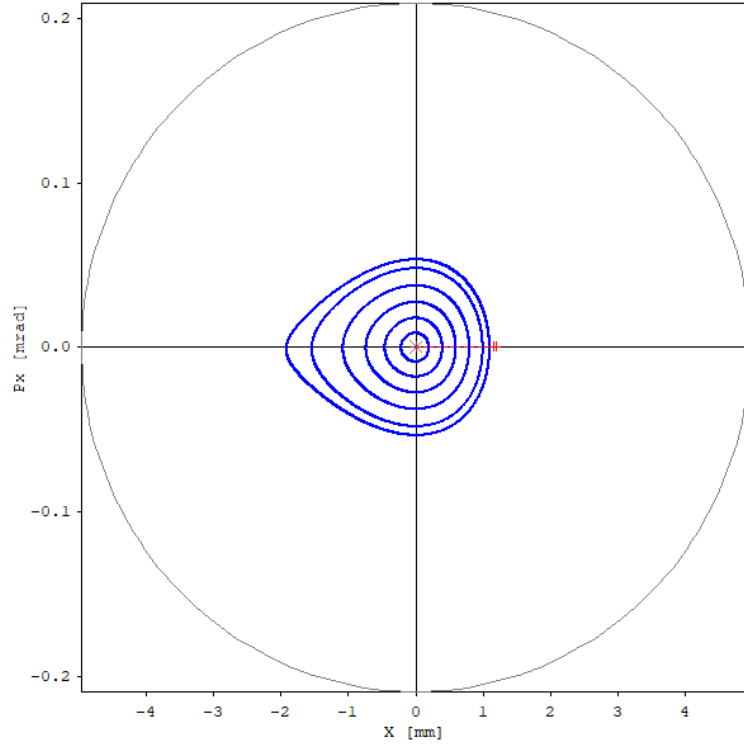


Figure 9: Horizontal phase-space tracking for the DBA lattice with sextupoles.

7.6. Harmonic Sextupole

The optimal location for a harmonic sextupole is a region where the horizontal dispersion is zero $D_x \approx 0$. With this approach, the sextupole does not contribute to chromaticity correction, since the chromatic effect is proportional to $\beta_x D_x$. Therefore, placing the harmonic sextupole in a dispersion-free region ensures that it does not modify the linear chromatic properties of the lattice.

While the chromatic sextupole families SH and SV are located in regions with non-zero dispersion to correct the natural chromaticity, the harmonic sextupole acts only in regions without dispersion so that particle position is independent of its energy deviation.

$$x = x_\beta + D_x \delta$$

Therefore, the harmonic sextupole only acts on the transverse motion and can cancel non-linear effects without changing the chromaticity, leading to an improvement of the dynamic aperture. For this reason, in the DBA cell represented in Figure 6 a reasonable choice is to place the

harmonic sextupole before the first bending or after the second bending of the DBA cell, where $D_x = 0$.

8. Conclusions

In this work, two different lattice designs have been studied and compared using OPA. First, a FODO ring was implemented and analysed. It was observed that QF mainly changes the horizontal tune, while QD is more efficient in the vertical plane. The independent tune control was also tested, showing that the analytical first-order formulas give a good approximation, although small differences appear because OPA does not use the thin-lens approximation.

Secondly, a DBA lattice was implemented. The QF2 strength was adjusted to suppress the dispersion at the ends of the cell, obtaining an achromatic solution. Then, the emittance was optimized and a much smaller horizontal emittance than in the FODO case was obtained, confirming that the DBA lattice is more efficient for reducing emittance. Finally, the natural chromaticity was calculated and corrected using two sextupole families placed where dispersion and beta functions are large. The chromaticity was corrected to zero, but the tracking showed that sextupoles introduce non-linear effects and reduce the dynamic aperture. Therefore, the final result shows that sextupoles are necessary to correct chromaticity, but they also reduce the stable region in phase space.